ML-Link: Limitations and Extensions

Link Prediction on Multilayer Networks

CS60078 Complex Networks — Project 10

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ML-Link: Summary

- Predicts missing links in **attributed multilayer networks**.
- Two complementary branches:
  - **GNN branch (ML-GAT)**: graph attention network on the supra-adjacency matrix.
  - **Structural branch**: explicit node-pair structural features.
    - ISL — Internal Structure Learning (within-layer overlap)
    - ESL — External Structure Learning (cross-layer)
    - CLA — Context-Level Attention
- Scores blended via a learned parameter λ; trained end-to-end with binary cross-entropy.

Identified Limitations

1. **1-hop structural overlap.** At high removal rates many node pairs share no 1-hop neighbours; structural score becomes identically zero. Baseline AUC drops from 97% at 50% removal to 64% at 90% removal.
2. **Identity matrix as node features.** GNN initialised with I_n — unique one-hot per node. Memorises which nodes are connected; no inductive generalisation.
3. **Static treatment of temporal networks.** Supra-adjacency aggregates all edges regardless of timestamp. Discards potentially informative temporal dynamics.
4. **Hyperparameter sensitivity (ψ).** Controls within- vs. cross-layer structural blend; requires re-tuning per dataset.

Limitations Summary

Limitation	**Consequence**	**Our response**
1-hop structural overlap	Structural score $= 0$ at high removal rates	k-hop extension (E2)
Identity matrix features	No inductive generalisation	Temporal side info (E6)
Static network assumption	Temporal dynamics ignored	Temporal encoder (E3–E6)
Hyperparameter sensitivity	Extra tuning per dataset	Fixed $\psi = 0.5$

Copenhagen Networks Study (CNS)

- $\approx$ 692 university students, 28 days, 4 interaction layers.
- Three layers carry timestamps; Facebook friendship is the static target.

Layer	**Index**	**Pairs**	**Density**	**Role**
Calls	0	697	0.3%	Context
SMS	1	697	0.3%	Context
Bluetooth Proximity	2	79,530	33.3%	Context
Facebook Friends	3	6,418	2.7%	Target

A Structural Mismatch

- **≈66% of fb_friends pairs have no recorded contact** in any context layer.
- Bluetooth measures physical co-location, not social closeness.
- Calls and SMS are informative but cover only 0.3% of pairs — too sparse.

Implication for temporal extensions:
A model relying on context-layer activity has nothing to work with for the majority of friendship pairs it must predict.

Temporal Preprocessing

- 28-day study divided into $T = 7$ windows ($\approx$4 days each).
- 9 normalised activity features per node per window:
  1. `calls_count_norm`, `calls_partners_norm`
  2. `sms_count_norm`, `sms_partners_norm`
  3. `bt_count_norm`, `bt_rssi_mean_norm`, `bt_partners_norm`
  4. `bt_active`, `calls_active` (binary)
- Activity tensor shape: $(692, 7, 9)$.

ML-Link Architecture

GNN Branch (ML-GAT)

- Multi-layer graph attention on supra-adjacency.
- Input features: identity matrix I_n.
- Functions as a learnable embedding table.

Structural Branch

- **ISL**: cosine similarity over within-layer 1-hop neighbours.
- **ESL**: cross-layer neighbourhoods (OAN, MAAN).
- **CLA**: attention weighting per layer.

- Final score: learned blend of both branches via λ.

Evaluation Protocol

- Edges removed from fb_friends at rates $r \in \{10\%, 20\%, \ldots, 90\%\}$.
- Remaining $(1 - r)$ fraction used for training; removed edges are positive test examples.
- Context layers (BT, calls, SMS) kept complete throughout.
- Metrics: **AUC** and **Average Precision (AP)**.
- Results reported as mean $\pm$ std over 5 independent runs.
- Negative evaluation set: uniformly sampled non-edges matching the positive count.

Approach 1: k-Hop Structural Features

Motivation: 1-hop overlap is zero for many pairs at high removal rates. Extending to k hops gives the structural branch more material.

Method: Per-layer k-hop adjacency with exponential decay on hop h:

$$A_l^{(k)} = \text{binarise}\left(A_l + \tfrac{1}{2}A_l^2 + \tfrac{1}{4}A_l^3 + \cdots + \tfrac{1}{2^{k-1}}A_l^k\right)$$

Binarised result replaces the 1-hop graph in ISL/ESL/CLA.
GNN branch unchanged — stacked GATv2 already has implicit k-hop receptive field.

At $k = 3$ on CNS: fb_friends 1-hop graph expands from ≈5,198 to ≈217,000 edges.

Approach 2: Temporal Side Information

Motivation: Three CNS layers carry timestamps. Can temporal dynamics predict friendship?

Temporal encoder maps $(N, 7, 9) \to (N, 64)$:

$$h_{\text{temp}} = \text{MeanPool}(\text{TransformerEncoder}(\text{Time2Vec}(\text{Linear}(x))))$$

- Time2Vec: learnable frequencies capture periodic patterns.
- Transformer: 2 layers, 4 heads, dropout 0.1.
- Pretrained 20 epochs on a self-supervised next-window activity prediction task.
- Pretraining loss converged to $\approx 0.69 \approx -\log 0.5$ — near random.

Integration of Temporal Side Information

Temporal embeddings **concatenated** with identity matrix:

$$\text{feats} = \left[I_n \;\middle|\; h_{\text{temp}} \right] \in \mathbb{R}^{n_{\text{layers}} \cdot N \times (N+64)}$$

- I_n retained: provides node distinguishability required by the GNN.
- h_{temp} adds complementary temporal context.
- Both tiled $n_{\text{layers}} = 4$ times to match supra-adjacency layout.

Removing I_n (E5) was explored but found to degrade performance sharply at higher removal rates.

Experiment Overview

ID	Description	Key change
E1	Baseline ML-Link	Standard model, identity features
E2	k-hop extension ($k = 3$)	Structural branch uses 3-hop adjacency
E3	Temporal branch, pair-level encoder	Separate scoring branch; blend via α
E4	Temporal branch + confident-wins	Disagreement resolved by higher-confidence score
E5	Side info, *no* identity matrix	h_{temp} replaces I_n
E6	Final: side info + identity matrix	$[I_n \mid h_{\text{temp}}]$ as node features

E1: Baseline ML-Link

- Standard ML-Link with identity matrix features and random negative sampling.
- AUC stable from 10%–60% removal (93–97%).
- Sharp drop at 80% and 90% (81% and 64%) — 1-hop structural overlap problem.
- High variance (std $\approx$5–7%) due to sensitivity to negative sampling.

Rate	10%	20%	30%	40%	50%
AUC	$93.05_{\pm 5.69}$	$96.85_{\pm 4.98}$	$96.55_{\pm 6.26}$	$96.14_{\pm 6.59}$	$97.10_{\pm 2.73}$

E2: k-Hop Extension ($k = 3$)

- Structural branch extended to 3-hop neighbourhoods.
- GNN branch deliberately **unchanged**.
- Gains of 2–3% over baseline at 30–50% removal.
- Variance reduced (std $\approx$1–3%).
- **Strongest result** in the study under fair evaluation conditions.

Rate	10%	20%	30%	40%	50%
AUC	$94.79_{\pm 4.00}$	$94.66_{\pm 3.98}$	$97.88_{\pm 2.87}$	$98.84_{\pm 1.30}$	$98.46_{\pm 1.24}$

E3 – E5: Temporal Variants

- **E3**: Pair-level temporal encoder as a separate scoring branch, blended via learned α.
  Marginally better at low removal; converges to baseline at higher rates.

- **E4**: Same as E3 but when the two branches disagree, the *more confident* prediction wins.
  No significant improvement.

- **E5**: Temporal embeddings replace I_n entirely.
  AUC appears high at low rates, but 64-dimensional embeddings are insufficient to distinguish 692 nodes. Performance degrades at higher removal rates.

E6: Final Design — Side Information with Identity Matrix

Architecture:

$$h_{\text{temp}} = \text{TemporalEncoder}(X_{\text{activity}}) \in \mathbb{R}^{N \times 64}$$

$$\text{feats} = [I_n \mid h_{\text{temp}}] \in \mathbb{R}^{n_{\text{layers}} \cdot N \times (N+64)}$$

$$\mathcal{L} = \text{BCE}(\text{score}_{\text{pos}}, \text{score}_{\text{neg}})$$

Rate	10%	20%	30%	40%
E6 AUC	$97.67_{\pm 1.24}$	$95.60_{\pm 2.54}$	$93.21_{\pm 3.47}$	$81.48_{\pm 5.34}$
Baseline AUC	$93.05_{\pm 5.69}$	$96.85_{\pm 4.98}$	$96.55_{\pm 6.26}$	$96.14_{\pm 6.59}$
ΔAUC	$+4.62$	-1.25	-3.34	-14.66

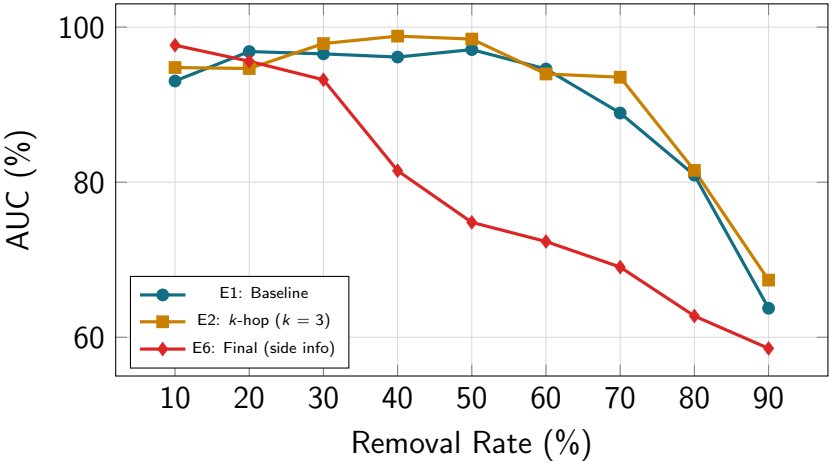
Degradation at $\geq 30\%$: sparse friendship labels insufficient to train the temporal encoder.

Full AUC Results — All Experiments

Exp.	Method	10%	20%	30%	40%	50%
E1	Baseline	93.05	96.85	96.55	96.14	97.10
E2	k-hop, $k = 3$	94.79	94.66	97.88	98.84	98.46
E3	Temporal v1	98.24	98.29	98.65	98.09	96.46
E4	Temporal v2 (rel. feats)	97.79	98.48	97.74	97.05	97.63
E5	Side info, no id. matrix	98.14	97.23	96.94	97.15	96.49
E6	Final (side info)	97.67	95.60	93.21	81.48	74.82

Green: genuine improvement over baseline. Red: degradation relative to baseline.

AUC vs. Removal Rate: E1, E2, E6



Why the k-Hop Extension Works

- Improvement most pronounced at **30–50% removal**.
- At these rates, roughly half of fb_friends edges are withheld — 1-hop common neighbours become scarce.
- 3-hop paths recover overlap that 1-hop misses, so ISL/MAAN/OAN return meaningful scores rather than zero.
- Richer representation also reduces sensitivity to the specific test-set sampling $\Rightarrow$ lower variance.

At 80–90% removal: even 3-hop paths are scarce; both models degrade substantially. Neither structural extension can compensate for the absence of graph signal.

Why Temporal Information Does Not Help

1. **Coverage gap**: ≈66% of friendship pairs have zero context-layer activity. Encoder outputs near-0.5 for these pairs — noise, not signal.
2. **Feature-label mismatch**: Bluetooth (the only dense layer, 33%) measures co-location, not friendship.
3. **Uninformative pretraining**: self-supervised loss converges to $\approx 0.69 \approx -\log 0.5$. Past activity is not predictive of future activity in CNS.
4. **Sparse supervision at high removal**: fewer than ≈4,000 positive training pairs at 30% removal — insufficient to fine-tune a Transformer encoder towards friendship.

The Role of the Identity Matrix

- I_n gives the GNN a unique n-dimensional signature for every node.
- Effectively a **learnable node embedding table** in the transductive setting.
- **E5**: removing I_n forced the model to distinguish 692 nodes with only 64-dimensional temporal embeddings — insufficient.
- Performance appeared adequate at low removal but degraded sharply at $\geq 30\%$.
- **Correct design (E6)**: retain I_n and concatenate temporal embeddings alongside it.

Key Implementation Issues

Issue	Symptom	Resolution
Identity matrix removed	AUC degrades at $\geq 30\%$	Restore I_n; concatenate with h_{temp}
`node_emb` variable scope	`UnboundLocalError` at checkpoint restore	Remove stale checkpoint code
`build_feats` signature mismatch	`TypeError` on function call	Align signature with current arguments
Node ID space mismatch	`IndexError` (index 799 out of range)	Apply same remapping as `cns_preprocess.py`
Relative temporal features	Performance below baseline at low removal	Revert to absolute activity features
`SparseEfficiencyWarning`	Warning from `scipy.sparse.setdiag`	Convert to `lil_matrix` before `setdiag`

Conclusion

1. ***k*-hop extension (E2)** is the only modification that produces a consistent, genuine improvement: **98.84% AUC at 40% removal** vs. 96.14% for the baseline.
2. **Temporal side information (E6)** degrades performance above 20% removal. The temporal encoder cannot learn useful representations when the signal is largely orthogonal to friendship.
3. **The identity matrix is essential** to ML-Link's GNN in the transductive setting and must be retained when adding supplementary features.

Future Directions

- **Temporal graph networks** (TGN, JODIE) operating on raw event streams rather than aggregated snapshots.
- **Causal temporal modelling**: time-aware train/test split to measure whether contact *precedes* friendship formation.
- **Generalisation**: evaluate k-hop extension on other datasets (Lazega, CKM, ArXiv) to confirm gains are not CNS-specific.
- **Systematic k sweep**: optimal k was fixed at 3; a principled selection criterion is left to future work.

Hyperparameters (E6)

Temporal encoder

Parameter	Value
d_{model}	64
Transformer layers	2
Attention heads	4
Dropout	0.1
Time windows T	7
Window length	$\approx$4 days
Pretraining epochs	20
Pretraining LR	10^{-3}

ML-Link & Training

Parameter	Value
Hidden dim	256
GNN layers	2
Attention heads	1
Dropout	0.5
Attn. dropout	0.7
ψ	0.5
Optimiser	Adam
LR	10^{-3}
Weight decay	10^{-5}
Max epochs	100
Patience	30
Runs	5

Full Results: AUC and AP

Exp.	Metric	10%	20%	30%	40%	50%	60%	70%	80%	90%
E1	AUC	93.05	96.85	96.55	96.14	97.10	94.60	88.92	80.91	63.74
	AP	93.01	96.80	96.71	96.37	97.29	94.87	89.60	81.79	66.51
E2	AUC	94.79	94.66	97.88	98.84	98.46	93.96	93.54	81.52	67.37
	AP	95.19	94.97	98.08	98.94	98.53	94.23	93.82	82.65	69.66
E3	AUC	98.24	98.29	98.65	98.09	96.46	94.19	89.36	80.61	62.83
	AP	98.36	98.33	98.73	98.23	96.58	94.47	89.88	81.57	64.77
E4	AUC	97.79	98.48	97.74	97.05	97.63	94.70	86.92	81.55	62.64
	AP	98.01	98.59	97.95	97.18	97.77	94.99	87.96	81.96	65.10
E5	AUC	98.14	97.23	96.94	97.15	96.49	93.74	88.46	79.82	62.73
	AP	98.19	97.17	96.99	97.26	96.61	94.11	89.26	81.06	64.79
E6	AUC	97.67	95.60	93.21	81.48	74.82	72.35	69.05	62.74	58.56
	AP	98.02	96.43	94.45	84.23	77.19	75.23	71.85	66.38	61.86

Thank You